

When you do:

```
import sklearn.datasets
```

`sklearn.datasets` provides **dataset loaders, fetchers, and generators**. Below is a **complete, practical list of datasets and dataset-related functions**, grouped by type (as in current stable scikit-learn versions).

◆ Toy / Small Built-in Datasets (`load_*`)

These return data immediately (no internet required).

Classification

- `load_iris`
- `load_digits`
- `load_wine`
- `load_breast_cancer`

Regression

- `load_diabetes`
- `load_linnerud`

Miscellaneous

- `load_sample_image`
 - `load_sample_images`
 - `load_files` (*generic loader for local text datasets*)
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◆ Real-World Datasets (Downloaded on Demand) (`fetch_*`)

Text / NLP

- `fetch_20newsgroups`
- `fetch_20newsgroups_vectorized`

Images

- `fetch_olivetti_faces`
- `fetch_lfw_people`

- `fetch_lfw_pairs`

Recommendation / Ratings

- `fetch_movielens`

Regression / Classification

- `fetch_california_housing`
- `fetch_covtype`
- `fetch_rcv1`
- `fetch_kddcup99`
- `fetch_openml`

◆ Synthetic Dataset Generators (`make_*`)

Used for testing, benchmarking, and experiments.

Classification

- `make_classification`
- `make_multilabel_classification`
- `make_blobs`
- `make_gaussian_quantiles`
- `make_hastie_10_2`
- `make_circles`
- `make_moons`

Regression

- `make_regression`
- `make_friedman1`
- `make_friedman2`
- `make_friedman3`

Clustering / Manifold

- `make_s_curve`
- `make_swiss_roll`

Matrix / Decomposition

- `make_low_rank_matrix`
- `make_sparse_coded_signal`

- `make_spd_matrix`
 - `make_sparse_spd_matrix`
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◆ Utilities & Helpers

- `clear_data_home`
 - `get_data_home`
 - `dump_svmlight_file`
 - `load_svmlight_file`
 - `load_svmlight_files`
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◆ Dataset Object

- `sklearn.datasets.base.Bunch` (*returned by most loaders*)
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◆ List everything programmatically

```
import sklearn.datasets as d
print([name for name in dir(d) if not name.startswith("_")])
```